Data-Driven Robust Performance Refinement of Dual-Stage Hard Disk Drive Controllers

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Objectives and Approach

Objective

- Design an **intuitive controller refinement algorithm** to optimize high-performance dual-stage hard disk drive (HDD) controllers.
- **Prove robust stability and performance** at every step of the algorithm.

Challenges

- **More systematic optimization** process versus indirect, experience-based, non-intuitive controller iterations.
- Representative metric of that assesses engineer's **true HDD control performance**.

Approach

- Utilize **black-box, zeroth-order (ZO) optimization algorithms** to push HDD performance.
- **Enforce robust stability** via a data-driven constraint derived from Cauchy's Argument Principle.

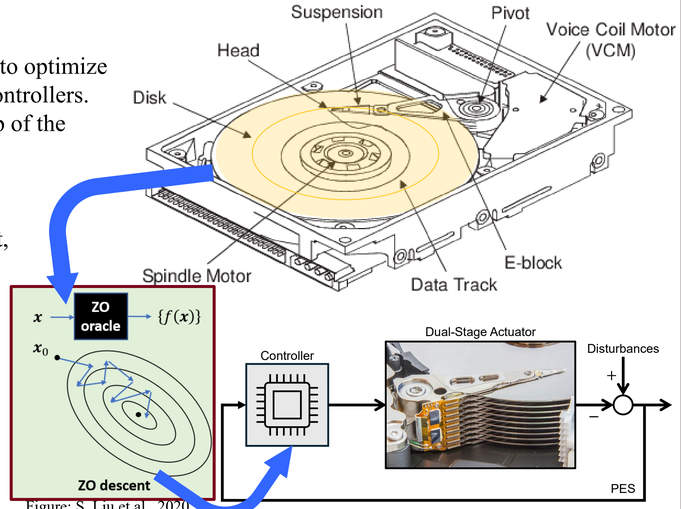
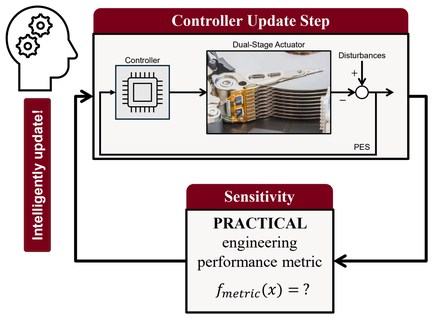


Figure: S. Liu et al., 2020

Photo: Oldham et al., 2008

Black-Box Approach



- **Traditional methods** push performance but have **conservative** robust constraints.
- **Proposed method** uses **quantitative metric** for the iterative design.
  - **Robust stability enforced**: must not iterate if a step is not stable!

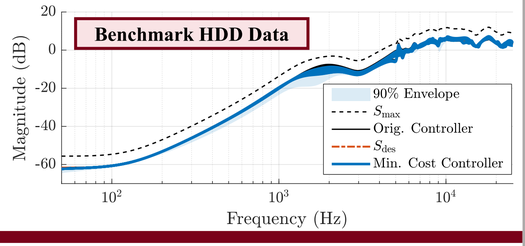
Controller Refinement Results

Test 1: No Objective, Minimize Sensitivity Response

Initial controller:
- Stable μ-syn controller

Objective Behavior:
- None, minimize baseline desired sensitivity

Algorithm Limits:
- 24-hour / 500,000 iteration limit

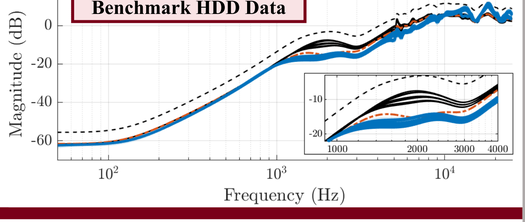


Test 2: Desired Sensitivity Notch at 2 kHz

Initial controller:
- Stable μ-syn controller

Objective Behavior:
- Desired sensitivity notch at 2 kHz w/ increased weighting

Algorithm Limits:
- 24-hour / 500,000 iteration limit

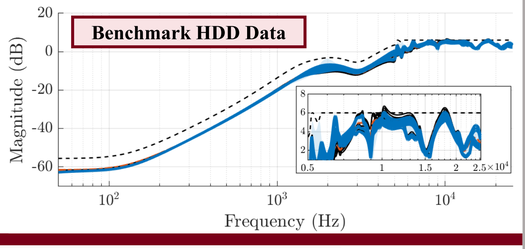


Test 3: Maximum Sensitivity Ceiling at 6 dB

Initial controller:
- Stable μ-syn controller

Objective Behavior:
- Max sensitivity ceiling at 6 dB

Algorithm Limits:
- 24-hour / 500,000 iteration limit

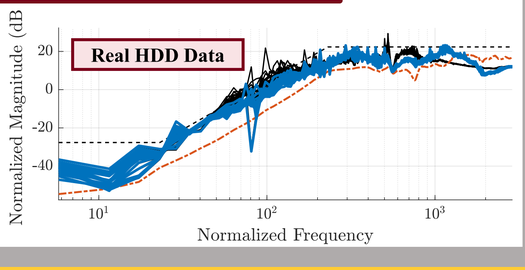


Test 4: Real Data Preliminary Aggressive Tests

Initial controller:
- Stable low-performance HDD controller

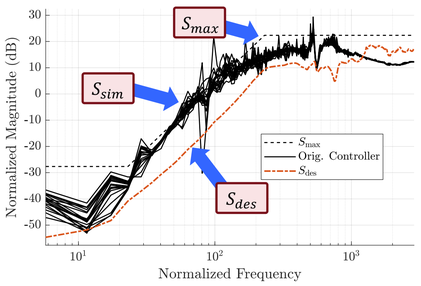
Objective Behavior:
- Max sensitivity high-pass, aggressive desired

Algorithm Limits:
- 12-hour / 60,000 iteration limit



Zeroth-Order (ZO) Optimization and Algorithm Logic

Performance (Cost) Function Design



- **Sensitivity** is a control engineer's **greatest metric** of HDD performance.
- **Performance cost function** that penalizes responses above a **desired sensitivity**.
- **Maximum sensitivity** soft constraint cost metric that **heavily penalizes violations.**

Performance Cost metric

$$Cost = f_p = \gamma_{scale} \sum_{\omega \in \Omega} \max(0, |S_{sim}|_{dB} - |S_{des}|_{dB})$$

Maximum Sensitivity Cost Metric

$$Cost = f_m = \gamma_{scale} \sum_{\omega \in \Omega} e^{\max(|S_{max}|_{dB} - |S_{sim}|_{dB})}$$

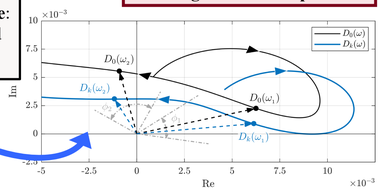
ZO Optimization Process

1. Our design vector to optimize
   $\theta_k = [\dots, \dots, \dots, \dots] \in \mathbb{R}^n$
2. Randomize a direction vector
   $\xi \in \mathbb{R}^n \qquad \xi \sim \mathcal{N}(0, I)$
3. Take an exploration step
   $\varepsilon \in \mathbb{R}$
   $\theta_{k,hi} = \theta_k + \varepsilon\xi$
   $\theta_{k,lo} = \theta_k - \varepsilon\xi$
   Central-point difference
4. Evaluate the steps
   $\nabla f(\theta_k) \approx \frac{f(\theta_{k,hi}) - f(\theta_{k,lo})}{2\varepsilon}$
   Directional gradient
5. Iterate on θ_k with α!
   $\theta_{k+1} = \theta_k - \alpha \nabla f(\theta_k)\xi$

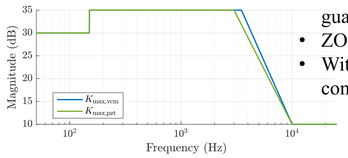
Data-Driven Robust Stability!

Inspired by current literature: model-based and data-driven!

Constraint based on Cauchy's Argument Principle:



Constraint on practical controller limits:



We use the best of current literature!
- Data-driven stability guarantee
- ZO optimization benefits
- Within feasible controller bounds

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